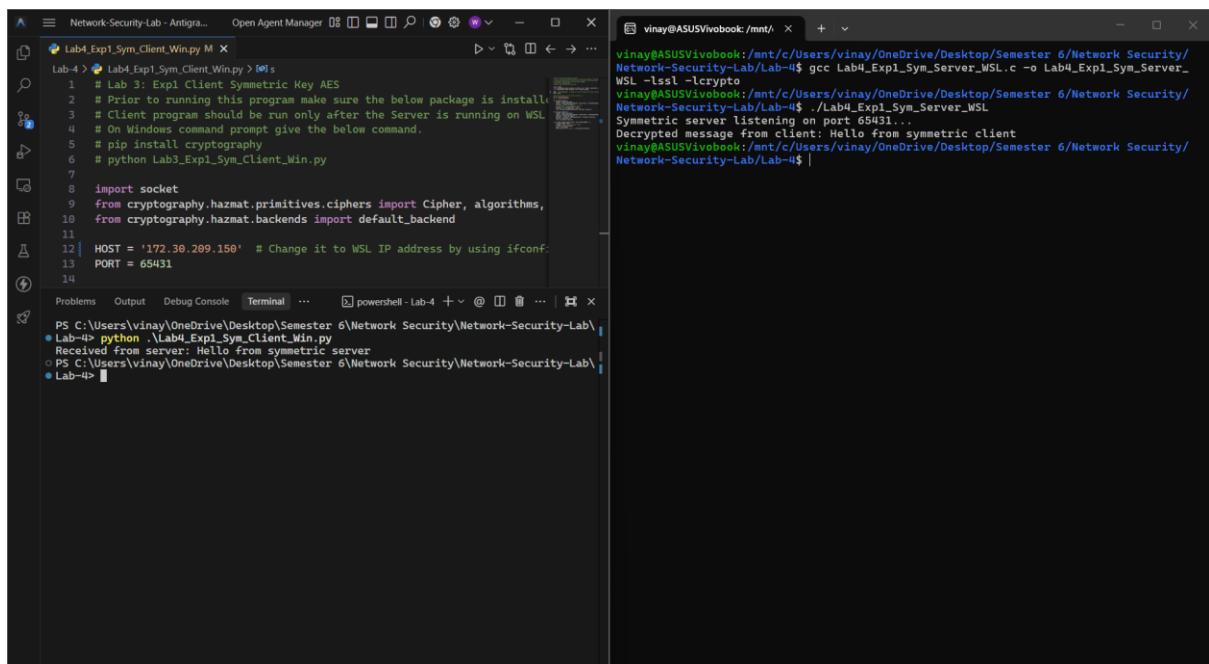


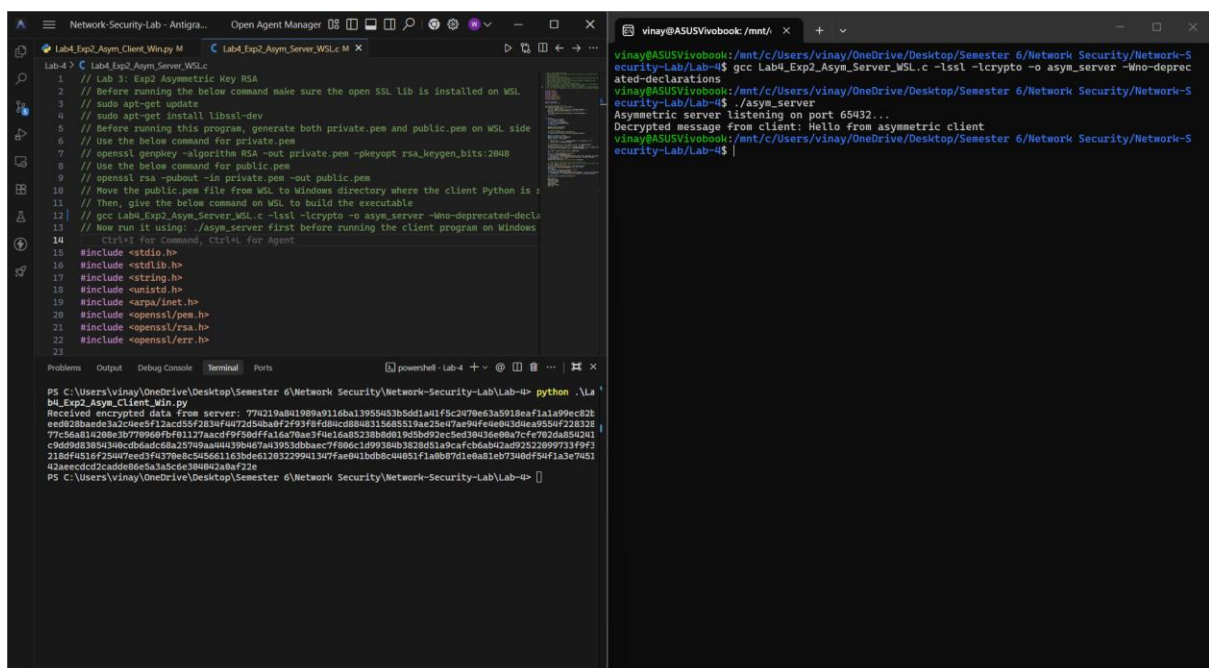
Lab-4-Exp-1 Client & Server



```
Lab4_Exp1_Sym_Client_Win.py M X
Lab-4 > Lab4_Exp1_Sym_Client_Win.py > 001 s
1 # Lab 3: Exp1 Client Symmetric Key AES
2 # Prior to running this program make sure the below package is install
3 # Client program should be run only after the Server is running on WSL
4 # On Windows command prompt give the below command.
5 # pip install cryptography
6 # python Lab3_Exp1_Sym_Client_Win.py
7
8 import socket
9 from cryptography.hazmat.primitives.ciphers import Cipher, algorithms,
10 from cryptography.hazmat.backends import default_backend
11
12 HOST = '172.30.209.150' # Change it to WSL IP address by using ifconf:
13 PORT = 65431
14
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab\
Lab-4> python .\Lab4_Exp1_Sym_Client_Win.py
Received from server: Hello from symmetric server
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab\
Lab-4>

vinay@ASUSVivobook: /mnt/ x + v
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/
Network-Security-Lab/Lab-4$ gcc Lab4_Exp1_Sym_Server_WSL.c -o Lab4_Exp1_Sym_Server_
WSL -lssl -lcrypto
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/
Network-Security-Lab/Lab-4$ ./Lab4_Exp1_Sym_Server_WSL
Symmetric server listening on port 65431...
Decrypted message from client: Hello from symmetric client
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/
Network-Security-Lab/Lab-4$ |
```

Lab-4-Exp-2 Client & Server



```
Lab4_Exp2_Asym_Client_Win.py M X Lab4_Exp2_Asym_Server_WSL.c M X
Lab-4 > Lab4_Exp2_Asym_Server_WSL.c
1 // Lab 3: Exp2 Asymmetric Key RSA
2 // Before running the below command make sure the open SSL lib is installed on WSL
3 // sudo apt-get update
4 // sudo apt-get install libssl-dev
5 // Before running this program, generate both private.pem and public.pem on WSL side
6 // Use the below command for private.pem
7 // openssl genpkey -algorithm RSA -out private.pem -pkeyopt rsa_keygen_bits:2048
8 // Use the below command for public.pem
9 // openssl rsa -pubout -in private.pem -out public.pem
10 // Move the public.pem file from WSL to Windows directory where the client Python is
11 // Then, give the below command on WSL to build the executable
12 // gcc Lab4_Exp2_Asym_Server_WSL.c -lssl -lcrypto -o asym_server -Wno-deprecated-decl
13 // Now run it using: ./asym_server first before running the client program on Windows
14 // Ctrl+C for Command, Ctrl+L for Agent
15 #include <stdio.h>
16 #include <stdlib.h>
17 #include <string.h>
18 #include <unistd.h>
19 #include <arpa/inet.h>
20 #include <openssl/pem.h>
21 #include <openssl/rsa.h>
22 #include <openssl/err.h>
23
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab\Lab-4> python .\La
b4_Exp2_Asym_Client_Win.py
Received encrypted data from server: 77a219a841989a9116ba13955453b5dd1a41f5c2470e63a5918a71a1a99ec82f
e0b28bae2a32c4ee5f12ac05f7283f407054baf7f93f6fd80c88a315685519ae25e47ae99fe0b43d4ea9554f228328
77c5e811280e307790609f8f117faacff9f806f1a167bae3fae1a8b23bb5801b5d5d92cc6ed39d36e0a7cf6782d4880241
c9dd9d98930c0dbdad6c8a2579aa44439b467a13953dbbaec7f806c1d99384b3828d51a9cafcb0ab42ad92522099733f9f3
218df4516f25a7eed3f4370e8c545661163bde61283229941347fae841bdb8c4051fa0b87d1e8a81eb7340df54f3e7a51
42aecc0f2c4de0e6a333ce030403a0bf22e
PS C:\Users\vinay\OneDrive\Desktop\Semester 6\Network Security\Network-Security-Lab\Lab-4>

vinay@ASUSVivobook: /mnt/ x + v
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-S
ecurity-Lab/Lab-4$ gcc Lab4_Exp2_Asym_Server_WSL.c -lssl -lcrypto -o asym_server -Wno-deprec
ated-declarations
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-S
ecurity-Lab/Lab-4$ ./asym_server
Asymmetric server listening on port 65432...
Decrypted message from client: Hello from asymmetric client
vinay@ASUSVivobook: /mnt/c/Users/vinay/OneDrive/Desktop/Semester 6/Network Security/Network-S
ecurity-Lab/Lab-4$ |
```